SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Pabshaliba S.V 1912260154 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Pabs
Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	5
Experimental Design	30	5
Use of Tools	20	5
Analysis of Results	15	5
Documentation	10	2

Total Marks 22 /30

Signature of the Faculty

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretation		
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



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6G Communication Architecture Compared with the OSI Model

Technical presentation

Presented by
1912260154-Paksha lika SV

Guided by
R Shankar

Introduction



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What is 6G Communication?

- 6G stands for Sixth-Generation Mobile Communication.
- It is the next generation of wireless communication after 5G.
- It aims to provide faster data transfer, lower latency, and intelligent connectivity.
- It is expected to support AI-powered networks, immersive experiences, and advanced sensing.
- The ITU identifies 6G as IMT-2030.

What is the OSI Model?

- OSI stands for Open Systems Interconnection.
- It is a seven-layer reference model for understanding network communication.
- It explains how data moves from one device to another



Objectives of the presentation

- Understand the basic concept of 6G communication.
- Study the proposed architecture of 6G networks.
- Understand the seven layers of the OSI model.
- Compare the functions of 6G architecture and the OSI model.
- Explore the potential applications and challenges of 6G.



Key features of 6G

- High data rates: Designed to support advanced, data-intensive applications.
- Low latency: Aims to reduce communication delays for time-sensitive tasks.
- AI integration: Uses artificial intelligence to help manage and optimize networks.
- Ubiquitous connectivity: Aims to connect people and devices across different environments.
- Integrated sensing: Communication infrastructure may also help detect and understand surrounding objects.

Proposed 6G communication architecture

A simplified conceptual 6G architecture can be represented using the following components:

Application layer: Smart healthcare, autonomous systems, immersive applications, and other user services.

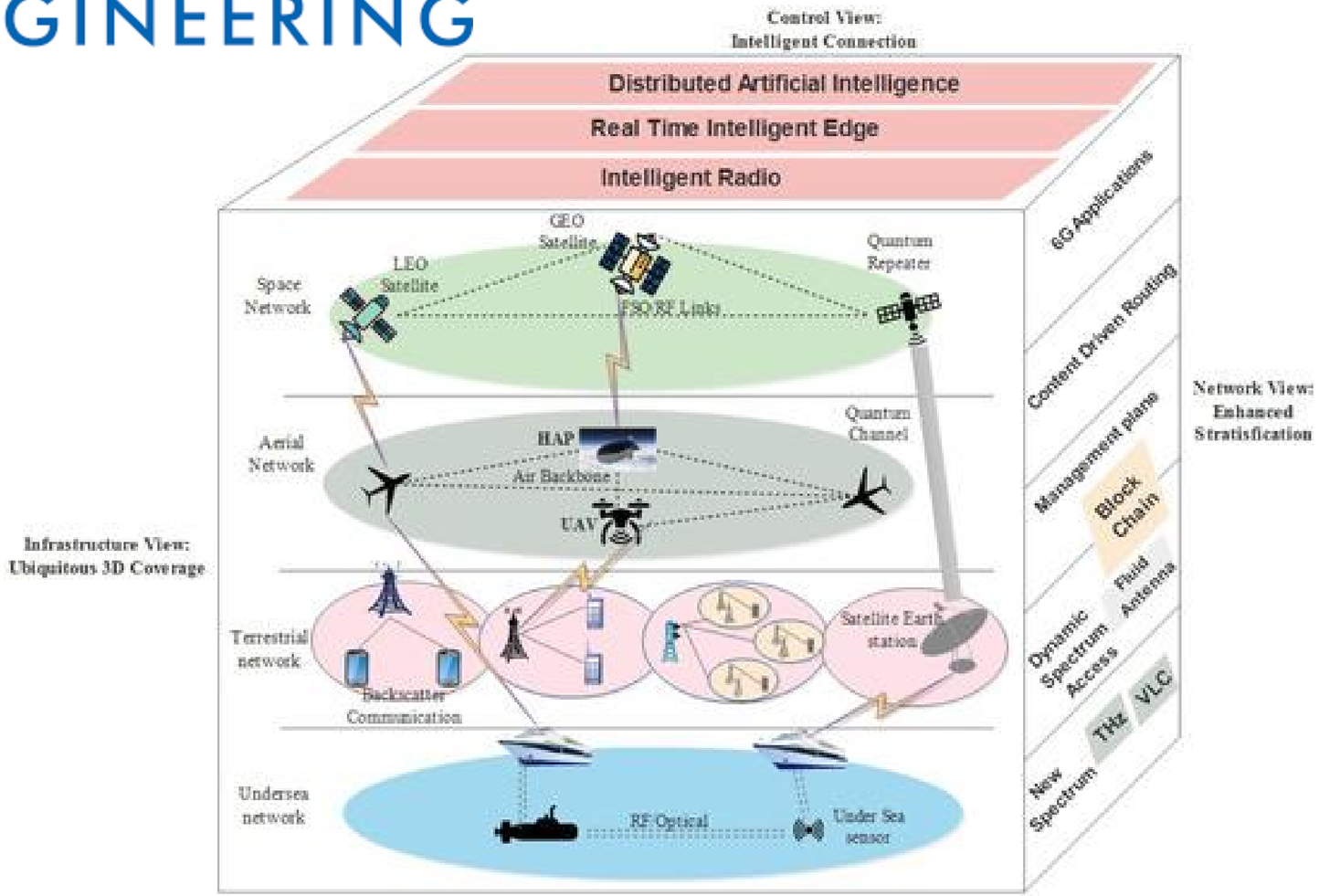
AI and service management: Helps optimize network resources and services.

Core network: Manages connectivity, mobility, authentication, and data routing.

Radio access network (RAN): Connects devices to the network through wireless access points.

Connectivity infrastructure: Includes terrestrial wireless links and potentially satellite or other non-terrestrial links.

User equipment: Smartphones, sensors, vehicles, wearables, and other connected devices





Components and technologies of 6G

- AI-native networking: Supports intelligent network management and resource allocation.
- Advanced antenna systems: Help improve wireless coverage and capacity.
- Terahertz and higher-frequency research: Investigates additional spectrum for high-capacity links.
- Edge computing: Processes data closer to users to reduce delays for suitable applications.
- Satellite-terrestrial integration: Can help extend connectivity to remote areas.
- Integrated sensing and communication (ISAC): Combines communication functions with sensing capabilities.

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Introduction to the OSI model

7	**Application**	Network services for user applications
6	**Presentation**	Data format, encryption and compression
5	**Session**	Establishes and manages communication sessions
4	**Transport**	End-to-end delivery and reliability
3	**Network**	Logical addressing and routing
2	**Data Link**	Frames and local link delivery
1	**Physical**	Transmits bits through a physical medium

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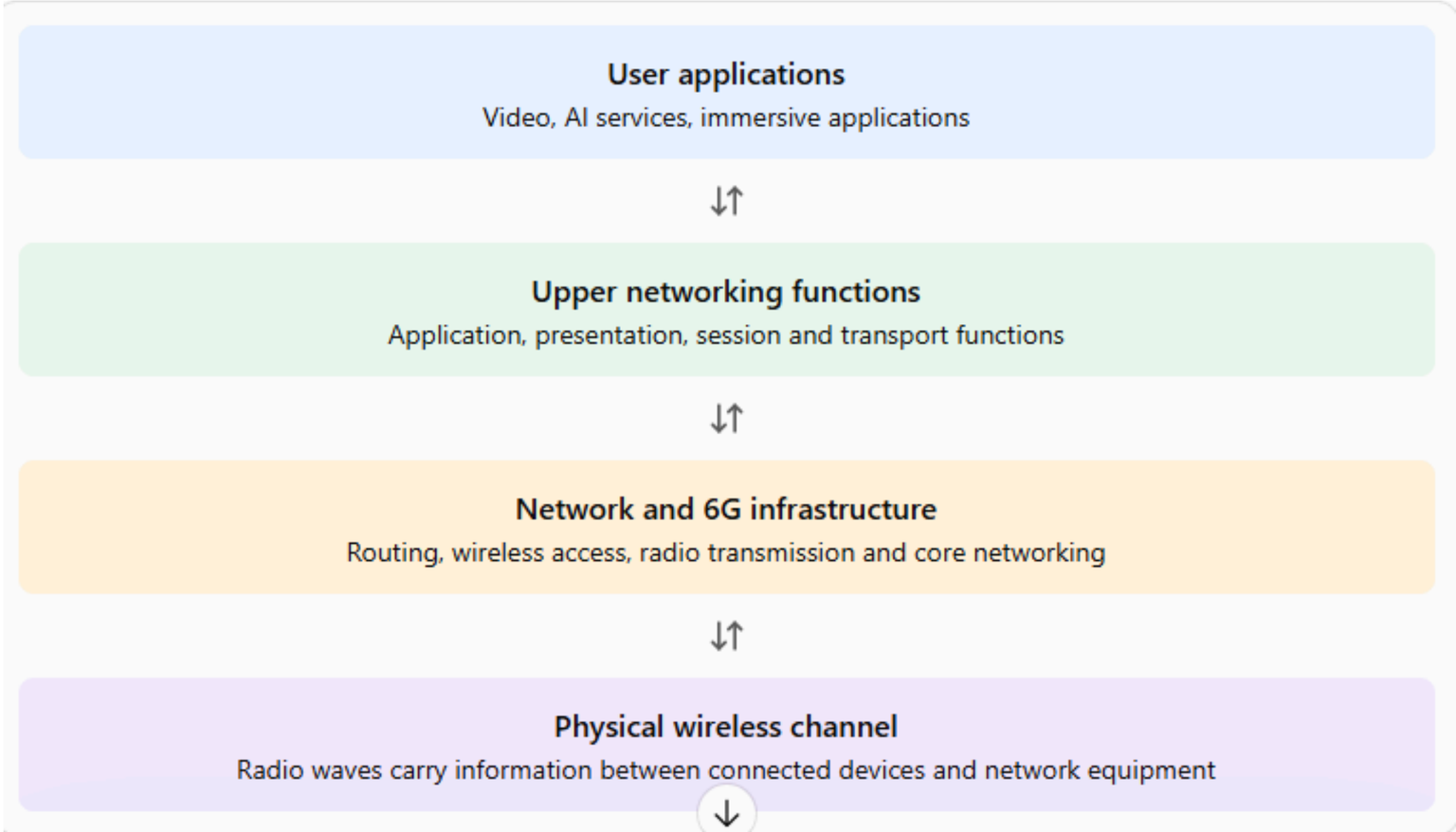
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Functions of the seven OSI layers

Layer	Main function	Example
Application	Provides network services	HTTP, DNS
Presentation	Data formatting and encryption	Data encoding, TLS-related functions
Session	Manages communication sessions	Session control
Transport	End-to-end data delivery	TCP, UDP
Network	Routing and logical addressing	IP
Data Link	Local delivery and framing	Ethernet, Wi-Fi MAC
Physical	Transmits bits	Radio signals, cables

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How 6G architecture and OSI work together



Comparison between 6G architecture and OSI model

Feature	6G communication architecture	OSI model
Type	Future mobile network architecture	Networking reference model
Main purpose	Provide wireless connectivity and services	Explain network communication functions
Structure	Devices, RAN, core, services and supporting infrastructure	Seven defined conceptual layers
Technology	AI integration, advanced radio and sensing	Technology-independent framework
Wireless communication	A central focus	Covered mainly by lower-layer concepts
Intelligence	AI integration is a key research direction	AI is not a dedicated OSI layer
Standardization	IMT-2030 development is ongoing	Established reference model

Main difference: 6G describes the network being developed; OSI describes how communication functions can be organized and understood.

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Conclusion and references

- 6G is the next generation of mobile communication, being developed under the IMT-2030 framework.
- Its research goals include intelligent connectivity, immersive communication, low latency, and integrated sensing.
- The OSI model divides networking into seven functional layers.
- The two concepts are complementary: 6G provides the communication infrastructure, while OSI helps explain the functions involved in transmitting data.
- Understanding both is useful for studying future wireless networks.

- **REFERNCE**

- International Telecommunication Union (ITU) — IMT-2030 framework and development.
- ITU — IMT-2030 technical requirements for 6G.
- Cisco — OSI reference model overview.

Thank you


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